EX:-4
Date:-
2/8/26

LAN using a switch and Ethernet cables.

AIM:-

Setup and configure a LAN (Local area network) using a switch & Ethernet cables in your lab.

What is a LAN?

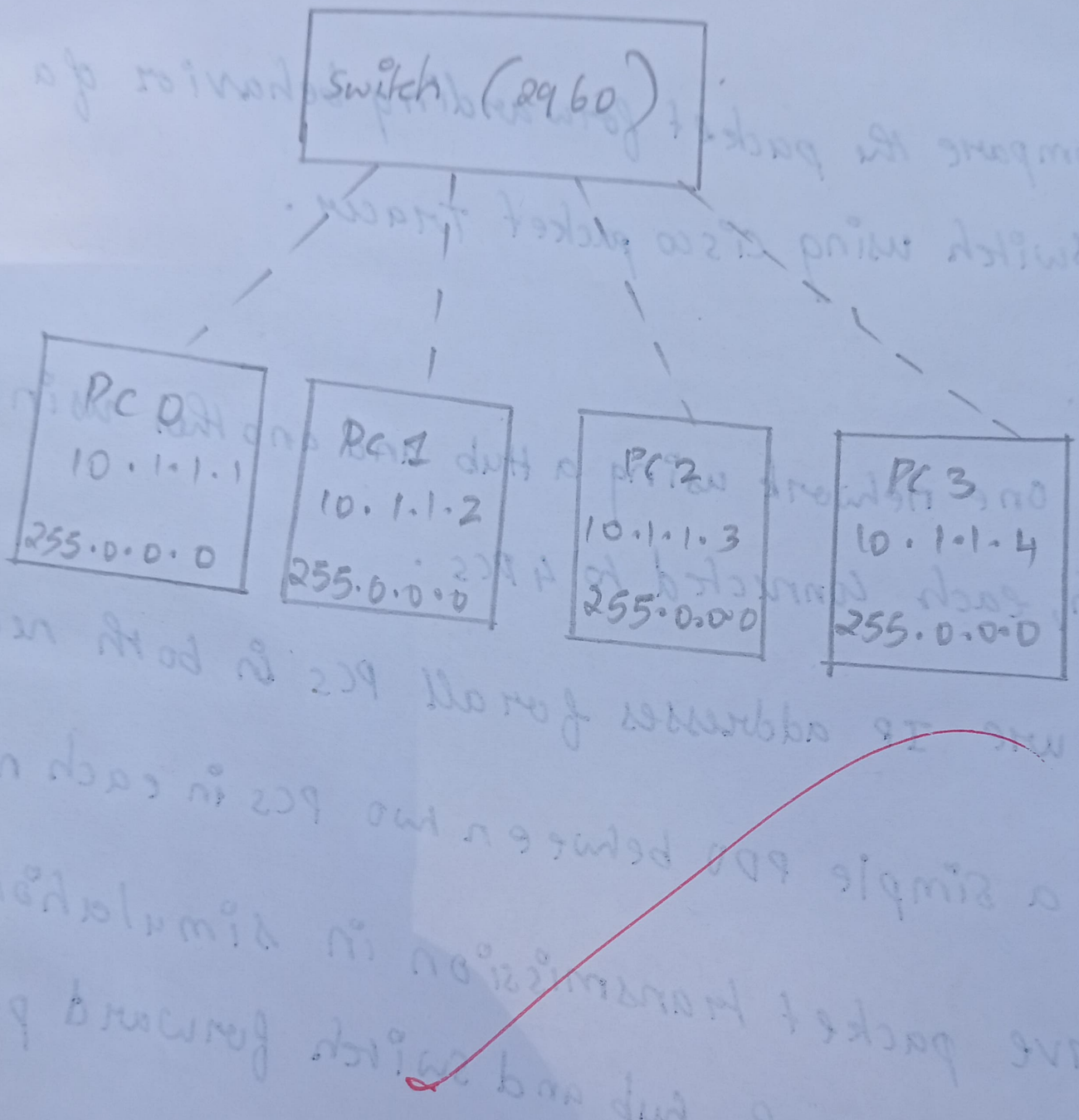
A local Area network (LAN) refers to a network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers, and internet information between users, such as computers, printers, servers, and switches. A local area network (LAN) switch serves as the primary connecting, device managing and directing communications within the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

Steps to set up a LAN:-

Step 1. Plan and Design an appropriate network topology taking into account network requirements and equipment location.

Step 2. Take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes, and 4 Ethernet cables.

Step 3: Connect the computers to network switch via an Ethernet cables, which is as simple as plugging one end of the Ethernet cables into the computer and the other end into the network switch.


Switch (2960)
PC 0
10.1.1.1
255.0.0.0
PC 1
10.1.1.2
255.0.0.0
PC 2
10.1.1.3
255.0.0.0
PC 3
10.1.1.4
255.0.0.0

Step 4 : Assign IP address to your PCs.

1) Log on to the client computer as Administrator or as owner.

2) Click Networks and Internet connections.

3) Right click local Area connection / Ethernet -> Go to properties -> select internet protocal (TCP/IPV4) -> click on properties -> select use the following ip address option and assign ipaddress.

Similarly assign IP address to all PCs connected to switch.

PC 1 - IPaddress: 10.1.1.1, Subnet mask 255.0.0.0

PC 2 - IPaddress - 10.1.1.2, Subnet mask 255.0.0.0

PC 3 - IP address - 10.1.1.3, Subnet mask 255.0.0.0

PC 4 - IPaddress 10.1.1.4, Subnet mask 255.0.0.0

Step 5 : Configure a network switch :

1) Connect the computer to the switch : To access the switch's web interface, it will need to connect the computer to the switch using an ethernet cable.

2) Log in to the web interface : open a web browser and enter the IP address of the switch in the address bar. This should bring up the login page for the switch's web interface. Enter the username and password to log in.

3) Configure basic settings : once logged in, it will be able to configure basic settings for the switch,

4) Assign IP address as: 10.1.1.5, subnet mask 255.0.0.0.

Step6:- check the connectivity between switch and other machine by using ping command in the command prompt of the device.

Step 7:- Select a folder -> go to properties -> click sharing tab -> share it with everyone on the same LAN.

Step8:- Try to access the shared folder from others computers of the network.

Result:-

A LAN was successfully configured using a switch and ethernet cables. IP addresses were assigned correctly. communication was verified the ping command, and file sharing was successfully demonstrated Computer LAN.